

LCC Projection Cone Apex Calculation

<https://gemini.google.com/app/6ae6c8906de8653d>

User prompt: LCC projection Standard Parallel 1 = 10°N Standard Parallel 2 = 30°N if R=8 cm , draw cross-section thru cone apex , North -south axis calculate center-earth to cone apex draw

Response: The distance from the center of the Earth to the cone's apex is **23.035 cm**.

Step-by-Step Calculation

To find the distance from the Earth's center to the apex of the secant cone, we treat this as a 2D coordinate geometry problem in the cross-section plane (the North-South axis).

Let the center of the Earth be the origin (0,0). The North-South polar axis is the y-axis, and the Equatorial plane is the x-axis.

1. Define the coordinates of intersection The cone intersects the Earth's surface (a circle of radius $R=8$ cm) at the two standard parallels:

- **Standard Parallel 1 ($\phi_1=10^\circ$):**
 $P_1=(R\cos 10^\circ, R\sin 10^\circ)$
- **Standard Parallel 2 ($\phi_2=30^\circ$):**
 $P_2=(R\cos 30^\circ, R\sin 30^\circ)$

2. Calculate the slope of the cone's side The line forming the side of the cone passes through P_1 and P_2 . Its slope (m) is:

$$m = \frac{R\cos 30^\circ - R\cos 10^\circ}{R\sin 30^\circ - R\sin 10^\circ} = \frac{\cos 30^\circ - \cos 10^\circ}{\sin 30^\circ - \sin 10^\circ}$$

Using standard sum-to-product trigonometric identities, this simplifies to:

$$m = -\cot(230^\circ + 10^\circ) = -\cot(20^\circ)$$

3. Find the y-intercept (the cone's apex) The apex sits on the y-axis ($x=0$). We find the y-intercept (y_{apex}) of the line $y - y_2 = m(x - x_2)$:

$$y_{\text{apex}} = R\sin 30^\circ - m(R\cos 30^\circ)$$

Substituting the slope and simplifying yields the general formula for the apex distance:

$$y_{apex} = R \sin(2\alpha + \phi) \cos(2\alpha - \phi)$$

4. Plug in the values

$$y_{apex} = 8 \sin(20^\circ) \cos(10^\circ)$$

$$y_{apex} = 80.34200.9848 \approx 8 \times 2.8794 \approx 23.035 \text{ cm}$$

Cross-Section Diagram

